Sahan S Liyanage

Experimental and Computational Physicist | Researcher | Scientific Software Developer

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Driven researcher with a strong background in experimental and computational physics, spanning astrophysics, spectroscopy, instrumentation, and scientific programming. Experienced in coordinating university courses and laboratories, publishing peer-reviewed work, presenting research, and supporting academic growth.

EDUCATION

Bachelor of Science Honours in Physics - Second Class (Upper Division)
Jan 2020 - May 2024

University of Colombo, Sri Lanka

- Cumulative GPA: 3.57/4.00 (126 credit hours).
- Relevant coursework included quantum mechanics, classical mechanics, electromagnetic fields, statistical physics, advanced optics, solid-state physics, nuclear and particle physics, computational physics, electronics, data acquisition, signal processing, and advanced physics laboratories.

Secondary Education
2015 - 2017

Central College, Anuradhapura, Sri Lanka

- GCE Advanced Level (2018), Physical Sciences: Combined Mathematics - B, Physics - B, Chemistry - B.

PROFESSIONAL EXPERIENCE

Research Assistant - Astronomy and Space Applications Division
May 2026 - Present

Arthur C. Clarke Institute for Modern Technologies, Katubedda, Moratuwa, Sri Lanka

- Conduct solar-radio observations with the e-CALLISTO antenna system, including calibration, performance checks, and hardware maintenance.
- Carry out H-alpha solar observations and record, process, and interpret the resulting images.
- Perform astronomical spectroscopy with optical telescopes and reduce and interpret the resulting spectra.
- Develop software and automated pipelines for observational data acquisition, processing, and archiving.
- Support public workshops and observation sessions in space physics and astronomy.

RESEARCH EXPERIENCE

Undergraduate Research Project
May 2023 - Apr 2024

Department of Physics, University of Colombo

Determination of the Speed and Source Height of Coronal Shock Waves Using Type II Solar Radio Bursts

- Analyzed 25 metric Type II solar-radio-burst datasets from the e-CALLISTO network.
- Developed a Python workflow for FITS reading, background-noise reduction, burst isolation, and dynamic-spectrum analysis.
- Derived shock-speed and source-height expressions using plasma-oscillation theory and the Newkirk coronal-density model.
- Estimated shock speeds of 343-1032 km/s and source heights of 1.317-1.724 solar radii.
- Supervisors: Dr. Janaka Adassuriya and Prof. K. P. S. C. Jayaratne.

Raman Spectroscopy Project
Nov 2022 - Mar 2023

Department of Physics, University of Colombo

Raman Spectroscopic Analysis of Common Minerals in Sri Lanka

- Analyzed calcite, dolomite, apatite, quartz, feldspar, graphite, and mica using a 532 nm Raman system.
- Prepared samples and processed spectra in Python using calibration, baseline correction, smoothing, and plotting.
- Supervisor: Prof. H. H. E. Jayaweera.

Undergraduate Research Mentor
May 2025 - Apr 2026

Final-year project mentoring

Estimating Coronal Magnetic Field Strength from Band Splitting in Type II Solar Radio Bursts

PROJECTS AND SCIENTIFIC SOFTWARE

Space Weather Monitoring Dashboard

2026 - Present

Full-stack solar and geospace monitoring application

- Combines e-CALLISTO spectra, GOES flux, SDO and SOHO imagery, Kp and Dst indices, and solar-wind and interplanetary magnetic-field data with alerts, a combined timeline, and archive search.
- ResNet-18 models achieved a test F1 score of 0.932 for burst detection and 0.966 validation accuracy with macro F1 0.889 for three-class burst typing.
- github.com/SaanDev/Solar_Monitoring_Dashboard

ecallistolib

2025 - Present

MIT-licensed Python library, version 1.3.1

- Downloads, reads, processes, combines, and plots e-CALLISTO FITS dynamic spectra with metadata-preserving parsing, RFI mitigation, background subtraction, and publication-quality output.
- Supports Python 3.10 through 3.14, a command-line interface, optional accelerated dependencies, and a pytest test suite.
- github.com/SaanDev/ecallistolib

e-CALLISTO FITS Analyzer

2024 - Present

Peer-reviewed, open-source Python desktop application

- Cross-platform framework for retrieving, cleaning, combining, visualizing, and analyzing e-CALLISTO dynamic spectra at event level.
- Uses Python, PySide6, NumPy, SciPy, Astropy, SunPy, Matplotlib, PyQtGraph, netCDF4, OpenPyXL, Requests, and BeautifulSoup.
- github.com/SaanDev/e-Callisto_FITS_Analyzer

Spectral Analysis of Fluids to Determine the Effects of Temperature

2021

General Physics Laboratory II, University of Colombo

- Used a home-made spectrometer and image-based absorbance measurements to estimate liquid temperature.
- Distilled water at 462 nm produced the best response: T = 73A + 230, with model errors of 4-12%.

PUBLICATIONS

Liyanage, G. L. S. S., Adassuriya, J., Jayaratne, K. P. S. C., Monstein, C., & Manoharan, P. K. (2026). e-CALLISTO FITS Analyzer: A software framework for CALLISTO solar radio data. *RAS Techniques and Instruments*, rzag056. doi:10.1093/rasti/rzag056

Liyanage, G. L. S. S., Adassuriya, J., Jayaratne, K. P. S. C., & Monstein, C. (2025). Determination of speed and source height of coronal shock waves using Type II solar radio bursts. *Proceedings of the Technical Sessions*, 41, 59-67. Institute of Physics, Sri Lanka. doi:10.48550/arXiv.2504.00622

SCIENTIFIC PRESENTATIONS

Determination of Speed and Source Height of Coronal Shock Waves Using Type II Solar Radio Bursts

22 Mar 2025

41st Technical Sessions, Institute of Physics, Sri Lanka - University of Colombo

- On-site oral presentation of undergraduate research findings.

Solar Radio Bursts and Space Weather

20 Nov 2024

Physics Society, University of Colombo

- On-site oral presentation for physics undergraduates.

Introduction to Solar Physics

18 Jul 2024

Astronomical Society, Rajarata University of Sri Lanka

- Online introductory lecture for university students.

TEACHING EXPERIENCE

Teaching Assistant

Aug 2025 - Mar 2026

University of Colombo School of Computing

- Probability and Statistics for first-year Computer Science majors.
- Digital Signal Processing for second-year Computer Science majors.
- Cryptography and Information Security for second-year Computer Science majors.

Teaching Assistant
May 2024 - May 2025

Department of Physics, University of Colombo

- Instructor in charge of Computational Electrodynamics and Advanced Physics Laboratory I.
- Instructor for Analogue and Digital Electronic Principles Laboratory.
- Instructor for General Physics Laboratory I and II.

ACADEMIC ENGAGEMENTS

Participant - COSPAR Capacity Building Workshop 2024
19-30 Aug 2024

Samarkand State University, Samarkand, Uzbekistan

- Practical training on coronal and interplanetary shocks using SOHO, SDO, STEREO, ground-based observations, SolarSoft, SunPy, AstroPy, and Python.

Designing Physics Laboratory Experiments

General Physics Laboratory II, Department of Physics, University of Colombo

- Designed the first-year optical-spectrometer laboratory sheet, including theory, pre-lab questions, prisms, refractive index, diffraction gratings, and atomic emission spectra.

SKILLS AND TECHNIQUES

Programming and scientific computing: Python, C/C++, MATLAB, SunPy, AstroPy, NumPy, SciPy, Matplotlib, FITS handling, signal processing, and Qt GUI development.

Software and tools: OriginPro, SolidWorks, Microsoft Office, Adobe Photoshop, LaTeX, Arduino, Linux, and web development.

Laboratory and instrumentation: Raman and UV-Vis spectroscopy, astronomical spectroscopy, e-CALLISTO radio spectrometry, analogue and digital electronics, data acquisition, signal processing, and microcontroller programming.

Management and leadership: Communication, project planning and execution, teaching, mentoring, and team coordination.

AWARDS AND HONOURS

Dean's List
Academic Year 2019/2020

University of Colombo

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

Vice President
2022 - 2023

Astronomical Society, University of Colombo

Coordinator
2020 - 2022

Astronomical Society, University of Colombo

Music Director
2022 - 2023

Music Society, University of Colombo

Marketing Director
2020 - 2021

Leo Club, University of Colombo

VOLUNTEERING EXPERIENCE

Science Bus - Phases I, II, and III
2021 - 2023

Physics Society, University of Colombo

- Contributed to an SPIE-funded travelling science project for school students in rural Sri Lanka.

Star Quest - All-Island Astronomical Quiz
2022 - 2025

Astronomical Society, University of Colombo

- Helped create questions, coordinate participants, run the event, and grade entries from school students across Sri Lanka.

Naada Nu 23 - Annual Musical Concert
2023

Music Society, University of Colombo

- Served as music director, contributed to event organization, and performed on stage.

REFERENCES

Professional references are available on request. Referee names, telephone numbers, and email addresses are intentionally omitted from this public web edition.